

Laboratory Record

submitted as part of

Programming for Data Science Lab (MACSE502)

LAB ASSESEMENT-5

by



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To



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School of Computer Science and Engineering



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

AUGUST 2026

2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record

INDEX

Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
1	09-07-2026	1	PP	Pandas	Basic Data Manipulation with Pandas	Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.	1
1	09-07-2026	2	EP	Pandas	AI Model Performance Analysis Using Pandas	An AI company maintains model performance records containing: • ModelID • Accuracy • Algorithm Perform: 1. Select ModelID and Accuracy. 2. Filter models with Accuracy greater than 90%. 3. Add Status with default value "Production Ready". 4. Sort models by Accuracy. 5. Calculate average model accuracy.	6
2	16-07-2026	3	PP	Pandas	Data Cleaning and Preprocessing with Pandas	Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling	14
2	16-07-2026	4	EP	Pandas	Movie Rating Data Preprocessing Using Pandas	A streaming platform maintains movie records containing: • MovieID • Rating • Genre Perform: 1. Create a dataset with missing values. 2. Fill missing Rating values using average rating. 3. Remove rows with missing MovieID or Genre. 4. Convert Rating into integer type. Normalize rating values.	18

3	23-07-2026	5	PP	NumPy	Creating and Manipulating Arrays Using NumPy	Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform the following operations: calculate the mean and standard deviation of the entire array, slice the array to get the first two rows and last two columns, reshape the array to shape (4, 3), and perform element-wise multiplication with another array of the same shape.	30
3	23-07-2026	6	EP	NumPy	Warehouse Shipment and Logistic Analytics Using Numpy	A logistics company tracks shipments from 4 warehouses across 7 weeks. Create a 4 x 7 shipment matrix. Perform the following operations: (1) compute average shipments, (2) compute standard deviation, (3) slice the first two warehouses and last three weeks, (4) reshape the matrix into a 7 x 4 matrix, and (5) perform element-wise multiplication with a transportation-cost matrix of the same shape. As an extra	35
4	30/7/2026	7	PP	Pandas, Numpy, Matplotlib	Feature Engineering and Exploratory Data Analysis	Create a DataFrame with columns Feature1, Feature2, and Target containing random data. Perform the following operations: create a new feature that is the logarithm of Feature1, bin Feature2 into three categories (low, medium, high), and calculate the correlation matrix of the DataFrame. Finally, create a scatter plot of Feature1 vs. Feature2 colored by Target, and interpret any visible patterns.	46

4	30/7/2026	8	EP	Pandas, NumPy, Matplotlib	Employee Attrition Prediction Using Workforce Indicators	A company wants to analyze workforce behavior and predict employee attrition. Create a DataFrame containing: OvertimeHours JobSatisfactionScore SalaryLevel TrainingHours WorkLifeBalance YearsAtCompany PromotionCount AttritionStatus Perform: 1. Create Log_OvertimeHours. 2. Categorize JobSatisfactionScore into: - o Disengaged - o Satisfied - o Highly Engaged 3. Generate correlation matrix. 4. Create a Treemap Visualization of employee categories. Analyze workforce trends.	54
5	06/08/2026	9	PP	Web Scraping	requests, BeautifulSoup, pandas, json	Scrape data from a webpage and store it in a structured format like CSV or JSON	63
5	06/08/2026	10	EP	Web Scraping	Online Job Market Analytics	A recruitment agency wants to analyze job opportunities posted on a dynamic job portal. Develop a Python application using Selenium to scrape job listings, extract Job Title, Company Name, Location and Salary (if available), store the collected data in a Pandas DataFrame, save the dataset as a CSV file, and visualize job demand by location.	70

PP: Practice Program; EP: Exercise Program; CP: Challenging Program

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Practice Program

Date: 06/08/2026

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Program S. No. 9

Title	Web Scraping and Data Collection			
Problem Statement	Scrape data from a webpage and store it in a structured format like CSV or JSON format			
Objectives	• To understand the concept of web scraping and data collection. • To retrieve webpage content using the Requests library. • To extract required information from HTML using BeautifulSoup. • To organize the scraped data using Pandas DataFrame. • To store the collected data in structured formats such as CSV and JSON.			
Dataset URL Link / File	https://quotes.toscrape.com			
Function Details	Sl. No.	Function	Package	Description of Function
	1	requests.get()	requests	Retrieves the HTML content of a webpage
	2	BeautifulSoup()	bs4	Parses the HTML content of the webpage
	3	find_all()	BeautifulSoup	Finds all matching HTML elements
	4	find()	BeautifulSoup	Finds the first matching HTML element
	5	DataFrame()	pandas	Converts collected data into tabular format
	6	to_csv()	pandas	Stores data in CSV format
	7	json.dump()	json	Stores Python data in JSON format
	8	json.load()	json	Reads data from a JSON file
Steps/ Process/ Procedure/	1. Import the required libraries: requests, BeautifulSoup, pandas and json. 2. Specify the URL of the webpage to be scraped.			

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Algorithm/ Flowchart/ Model	3. Send an HTTP GET request using requests.get(). 4. Parse the HTML content using BeautifulSoup. 5. Locate the required HTML elements containing quotes, authors and tags. 6. Extract the required data from the webpage. 7. Store the extracted data in a Python list. 8. Convert the collected data into a Pandas DataFrame. 9. Save the structured data in CSV format using to_csv(). 10. Save the same data in JSON format using json.dump(). 11. Display the collected data to verify the output.
Program	<pre> import requests from bs4 import BeautifulSoup import pandas as pd import json url = "https://quotes.toscrape.com/" response = requests.get(url) print("Status Code:", response.status_code) soup = BeautifulSoup(response.text, "html.parser") print("Website Title:", soup.title.text) quotes = [] for item in soup.find_all("div", class_="quote"): quote = item.find("span", class_="text").text author = item.find("small", class_="author").text tags = [tag.text for tag in item.find_all("a", class_="tag")] quotes.append({ "Quote": quote, "Author": author, "Tags": " ".join(tags) }) print("Number of quotes collected:", len(quotes)) </pre>

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```
df = pd.DataFrame(quotes)

df

df.to_csv("quotes.csv", index=False)

print("Data saved successfully in CSV format.")

with open("quotes.json", "w", encoding="utf-8") as file:
    json.dump(quotes, file, indent=4, ensure_ascii=False)

print("Data saved successfully in JSON format.")

csv_data = pd.read_csv("quotes.csv")

csv_data

all_quotes = []

page = 1

while True:

    url = f"https://quotes.toscrape.com/page/{page}/"

    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")

    items = soup.find_all("div", class_="quote")

    if not items:
        break

    for item in items:

        quote = item.find("span", class_="text").text
        author = item.find("small", class_="author").text

        tags = [
            tag.text
            for tag in item.find_all("a", class_="tag")
        ]

        all_quotes.append({
            "Quote": quote,
```

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	<pre> "Author": author, "Tags": ", ".join(tags) }) page += 1 print("Total quotes collected:", len(all_quotes)) df = pd.DataFrame(all_quotes) df df.to_csv("quotes_dataset.csv", index=False) with open("quotes_dataset.json", "w", encoding="utf-8") as file: json.dump(all_quotes, file, indent=4, ensure_ascii=False) print("CSV and JSON files created successfully.") </pre>
Output Snapshots With Explanation/ Observations	The required libraries requests, BeautifulSoup, pandas and json are imported for web scraping, HTML parsing, data organization and structured data storage. <pre> 1 import requests 2 from bs4 import BeautifulSoup 3 import pandas as pd 4 import json </pre> <pre> 1 url = "https://quotes.toscrape.com/" 2 3 response = requests.get(url) 4 5 print("Status Code:", response.status_code) </pre> Status Code: 200 <i>Fig. 1.1: Importing Required Python Libraries</i>

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The webpage <https://quotes.toscrape.com/> is accessed using `requests.get()`. The status code 200 confirms that the webpage was successfully retrieved.

```
[7]: 1 quotes = []
      2
      3 for item in soup.find_all("div", class_="quote"):
      4
      5     quote = item.find("span", class_="text").text
      6     author = item.find("small", class_="author").text
      7
      8     tags = [
      9         tag.text
     10         for tag in item.find_all("a", class_="tag")
     11     ]
     12
     13 quotes.append({
     14     "Quote": quote,
     15     "Author": author,
     16     "Tags": ", ".join(tags)
     17 })
     18
     19 print("Number of quotes collected:", len(quotes))

Number of quotes collected: 10
```

Fig. 1.2: Accessing the Webpage

The HTML content of the webpage is parsed using BeautifulSoup. The webpage title confirms that the correct page has been successfully loaded.

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```
number of quotes collected: 10

[8]: 1 df = pd.DataFrame(quotes)
      2
      3 df

[8]:
```

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig. 1.3: Parsing the HTML and displaying the dataset Content

```
1 df.to_csv("quotes.csv", index=False)
2
3 print("Data saved successfully in CSV format.")

Data saved successfully in CSV format.
```

Fig. 1.4: Saving Data in CSV Format

The scraped data is successfully stored in CSV format using the pandas `to_csv()` function.

The scraped data is successfully stored in JSON format using the `json.dump()` function.

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```
1 with open("quotes.json", "w", encoding="utf-8") as file:
2     json.dump(quotes, file, indent=4, ensure_ascii=False)
3
4 print("Data saved successfully in JSON format.")
```

Data saved successfully in JSON format.

Fig. 1.5: Saving Data in JSON Format

	Quote	Author	Tags
0	"The world as we have created it is a process ...	Albert Einstein	change, deep-thoughts, thinking, world
1	"It is our choices, Harry, that show what we t...	J.K. Rowling	abilities, choices
2	"There are only two ways to live your life. On...	Albert Einstein	inspirational, life, live, miracle, miracles
3	"The person, be it gentleman or lady, who has ...	Jane Austen	aliteracy, books, classic, humor
4	"Imperfection is beauty, madness is genius and...	Marilyn Monroe	be-yourself, inspirational
5	"Try not to become a man of success. Rather be...	Albert Einstein	adulthood, success, value
6	"It is better to be hated for what you are tha...	André Gide	life, love
7	"I have not failed. I've just found 10,000 way...	Thomas A. Edison	edison, failure, inspirational, paraphrased
8	"A woman is like a tea bag; you never know how...	Eleanor Roosevelt	misattributed-eleanor-roosevelt
9	"A day without sunshine is like, you know, nig...	Steve Martin	humor, obvious, simile

Fig 1.6: Displaying the saved json and csv

The scraped data is successfully stored in JSON format using the `json.dump()` function.

The saved CSV file is read using `pandas.read_csv()` and displayed to verify that the data has been stored correctly.

The saved JSON file is read using `json.load()` and the stored records are displayed successfully, confirming correct JSON storage.

Conclusions

The experiment was successfully completed by scraping data from the given webpage using Python. The webpage content was retrieved using the requests library and parsed using BeautifulSoup. The required information such as quotes, authors, and tags was extracted and organized using a Pandas DataFrame. Finally, the collected data was successfully stored in structured CSV and JSON formats for further analysis and use.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data
2. Python Program: [GitHub link](#)
3. Class lecture: [Lecture link](#)

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Program S. No. 10

Title	Online Job Market Analytics			
Problem Statement	A recruitment agency wants to analyze job opportunities posted on a dynamic job portal. Develop a Python application using Selenium to scrape job listings, extract Job Title, Company Name, Location and Salary (if available), store the collected data in a Pandas DataFrame, save the dataset as a CSV file, and visualize job demand by location.			
Objectives	• To understand web scraping from a dynamic job portal using Selenium. • To extract job title, company name, location and salary information from job listings. • To organize and clean the collected data using a Pandas DataFrame. • To save the structured job dataset in CSV format. • To perform location-wise analysis and visualize job distribution using Matplotlib.			
Dataset URL Link / File	Job Portal: https://www.naukri.com Search role used: Data Scientist Cities: Bangalore, Hyderabad, Pune, Chennai, Mumbai and Delhi			
Functions Details	Sl. No.	Function	Package	Description of function
	1	Firefox()	selenium	Starts a Firefox browser session controlled by Selenium.
	2	driver.get()	selenium	Opens the specified job-portal URL in the automated browser.
	3	execute_script()	selenium	Executes JavaScript to scroll the dynamically loaded webpage.
	4	find_elements()	selenium	Finds all matching job-card elements on the page.
	5	find_element()	selenium	Finds a required field inside an individual job card.
	6	time.sleep()	time	Pauses execution briefly to allow dynamic content to load.
	7	DataFrame()	pandas	Converts the collected job records into tabular form.
	8	drop_duplicates()	pandas	Removes duplicate job records from the DataFrame.
	9	reset_index()	pandas	Resets row indexing after data cleaning.
	10	to_csv()	pandas	Saves the structured job dataset as a CSV file.
	11	value_counts()	pandas	Counts job records for each location for analysis.
	12	bar()/savefig()	matplotlib	Creates and saves the location-wise job-demand bar chart.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import Selenium, Pandas, Matplotlib and supporting Python modules. 2. Define the job role, location and maximum number of job listings to collect. 3. Launch the browser using Selenium WebDriver and construct the Naukri search URL. 4. Open the dynamic job-search page and allow sufficient time for the page to load. 5. Scroll the webpage so that dynamically loaded job cards become available. 6. Locate job-card elements using CSS selectors. 7. Extract Job Title, Company Name, Location and Salary from each job card. 8. Store the extracted records in a Python list and convert them into a Pandas DataFrame. 9. Remove duplicate records and save the cleaned dataset as a CSV file. 10. Repeat the search for Bangalore, Hyderabad, Pune, Chennai, Mumbai			

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	and Delhi. 11. Use value_counts() to prepare location-wise job analysis and create a bar chart. 12. Save the visualization and multi-city CSV file, display the final summary and close the browser. START -> Open Job Portal -> Scrape Listings -> DataFrame -> CSV -> Location Analysis -> Bar Chart -> STOP
Program	<pre> import time import pandas as pd import matplotlib.pyplot as plt from selenium import webdriver from selenium.webdriver.common.by import By from selenium.webdriver.firefox.options import Options from selenium.common.exceptions import NoSuchElementException JOB_ROLE = "data scientist" LOCATION = "bangalore" MAX_JOBS = 30 options = Options() driver = webdriver.Firefox(options=options) driver.set_window_size(1400, 900) def make_slug(text): return "-".join(text.lower().strip().split()) def get_text(card, selectors, default="Not Available"): for selector in selectors: try: element = card.find_element(By.CSS_SELECTOR, selector) text = element.text.strip() if text: return text except NoSuchElementException: pass return default job_slug = make_slug(JOB_ROLE) location_slug = make_slug(LOCATION) url = f"https://www.naukri.com/{job_slug}-jobs-in-{location_slug}" print("Search URL:", url) driver.get(url) time.sleep(5) print("Page Title:", driver.title) print("Current URL:", driver.current_url) </pre>

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```
for i in range(3):
    driver.execute_script("window.scrollTo(0,
document.body.scrollHeight);")
    time.sleep(2)
driver.execute_script("window.scrollTo(0, 0);")

cards = driver.find_elements(By.CSS_SELECTOR, "div.cust-job-tuple")
print("Number of job cards found:", len(cards))

jobs = []
for card in cards[:MAX_JOBS]:
    title = get_text(card, ["a.title", "[class*='title']"])
    company = get_text(card, ["a.comp-name", ".comp-name"])
    location = get_text(card, ["span.locWdth", ".locWdth",
"[class*='location']"])
    salary = get_text(card, ["span.sal", ".sal", "[class*='salary']"],
default="Not Disclosed")
    jobs.append({
        "Job Title": title,
        "Company Name": company,
        "Location": location,
        "Salary": salary
    })

print("Jobs scraped:", len(jobs))
df = pd.DataFrame(jobs)
df = df.drop_duplicates().reset_index(drop=True)
print("Total unique jobs:", len(df))
print("Dataset Shape:", df.shape)
print(df.head(10))
print("Salary Information:")
print(df["Salary"].value_counts())

df.to_csv("job_market_data.csv", index=False)
print("CSV file saved successfully.")

csv_data = pd.read_csv("job_market_data.csv")
print(csv_data.head())

cities = ["bangalore", "hyderabad", "pune", "chennai", "mumbai",
"delhi"]
all_jobs = []

for city in cities:
    city_slug = make_slug(city)
    url = f"https://www.naukri.com/{job_slug}-jobs-in-{city_slug}"
    print("Scraping:", city)
```

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```
driver.get(url)

time.sleep(4)

city_cards = driver.find_elements(By.CSS_SELECTOR, "div.cust-job-  
tuple")
print("Jobs found:", len(city_cards))

for card in city_cards[:10]:
    title = get_text(card, ["a.title", "[class*='title']"])
    company = get_text(card, ["a.comp-name", ".comp-name"])
    salary = get_text(card, ["span.sal", ".sal", "[class*='salary']"],  
                        default="Not Disclosed")
    all_jobs.append({
        "Job Title": title,
        "Company Name": company,
        "Location": city.title(),
        "Salary": salary
    })

city_df = pd.DataFrame(all_jobs)
city_df = city_df.drop_duplicates().reset_index(drop=True)
print("Total jobs collected:", len(city_df))

location_analysis = city_df["Location"].value_counts().reset_index()
location_analysis.columns = ["Location", "Number of Jobs"]
print(location_analysis)

plt.figure(figsize=(10, 5))
plt.bar(location_analysis["Location"], location_analysis["Number of  
Jobs"])
plt.title("Data Scientist Job Demand by Location")
plt.xlabel("Location")
plt.ylabel("Number of Job Listings")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("job_demand_by_location.png", dpi=300,  
bbox_inches="tight")
plt.show()

city_df.to_csv("job_market_multi_city.csv", index=False)
print("Multi-city dataset saved successfully.")

print("ONLINE JOB MARKET ANALYTICS")
print("Job Role:", JOB_ROLE)
print("Total Jobs Analyzed:", len(city_df))
print("Cities Analyzed:", len(location_analysis))
```

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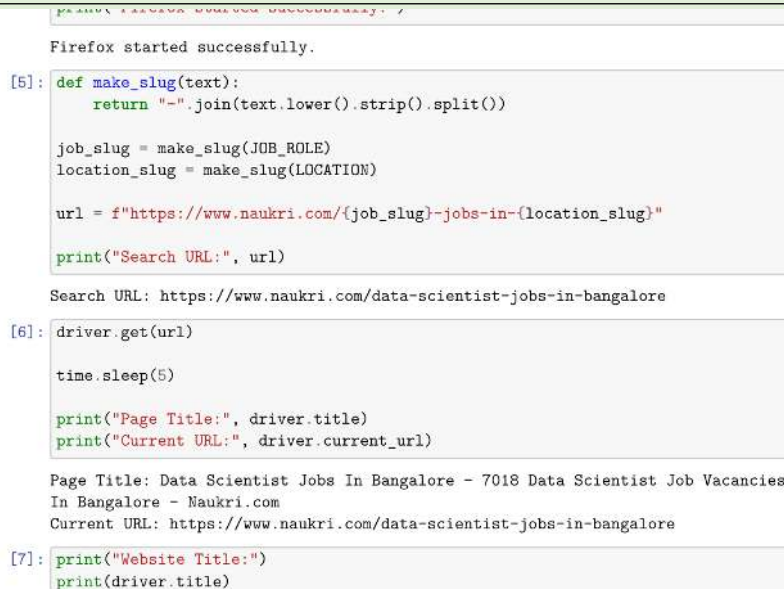
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Program S. No. 10

**Output
Snapshots
With
Explanation/
Observations**

```
print(location_analysis.to_string(index=False))

driver.quit()
print("Browser closed successfully.")
```



```

Python 3.10.12 Shell - MACSE502 - 7
Firefox started successfully.
[5]: def make_slug(text):
      return "-".join(text.lower().strip().split())

      job_slug = make_slug(JOB_ROLE)
      location_slug = make_slug(LOCATION)

      url = f"https://www.naukri.com/{job_slug}-jobs-in-{location_slug}"
      print("Search URL:", url)

      Search URL: https://www.naukri.com/data-scientist-jobs-in-bangalore
[6]: driver.get(url)

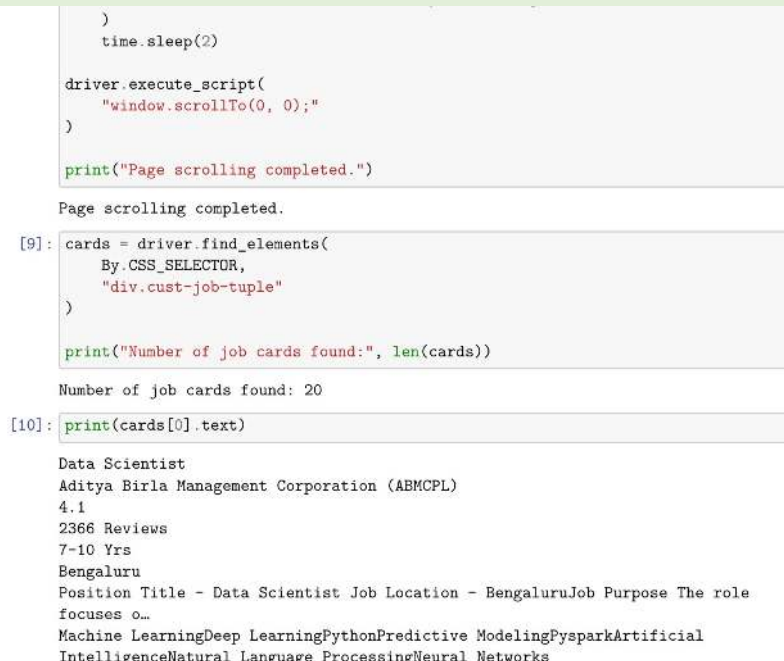
      time.sleep(5)

      print("Page Title:", driver.title)
      print("Current URL:", driver.current_url)

      Page Title: Data Scientist Jobs In Bangalore - 7018 Data Scientist Job Vacancies
      In Bangalore - Naukri.com
      Current URL: https://www.naukri.com/data-scientist-jobs-in-bangalore
[7]: print("Website Title:")
      print(driver.title)
  
```

Fig. 1.1: Search URL and Job Portal Validation

The Selenium session generated the Naukri search URL for Data Scientist jobs in Bangalore and successfully retrieved the page title and current URL, confirming that the dynamic job portal was opened correctly.



```

      )
      time.sleep(2)

      driver.execute_script(
          "window.scrollTo(0, 0);"
      )

      print("Page scrolling completed.")

      Page scrolling completed.
[9]: cards = driver.find_elements(
      By.CSS_SELECTOR,
      "div.cust-job-tuple"
      )

      print("Number of job cards found:", len(cards))

      Number of job cards found: 20
[10]: print(cards[0].text)

      Data Scientist
      Aditya Birla Management Corporation (ABMCPL)
      4.1
      2366 Reviews
      7-10 Yrs
      Bengaluru
      Position Title - Data Scientist Job Location - BengaluruJob Purpose The role
      focuses o...
      Machine LearningDeep LearningPythonPredictive ModelingPysparkArtificial
      IntelligenceNatural Language ProcessingNeural Networks
  
```

Fig. 1.2: Detection of Dynamic Job Cards

After scrolling the page, Selenium detected 20 job-card elements. The

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first job card displays details such as job title, company, experience, location and associated skills.

```
jobs.append({
    "Job Title": title,
    "Company Name": company,
    "Location": location,
    "Salary": salary
})

print("Jobs scraped:", len(jobs))

Jobs scraped: 20

[14]: jobs[:3]

[14]: [{'Job Title': 'Data Scientist',
      'Company Name': 'Aditya Birla Management Corporation (ABMCPL)',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist/Software Developer',
      'Company Name': 'Xylem Water Solutions',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'},
      {'Job Title': 'Data Scientist III',
      'Company Name': 'Walmart',
      'Location': 'Bengaluru',
      'Salary': 'Not Disclosed'}]

[15]: df = pd.DataFrame(jobs)

df

[15]:
```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG
5	Machine Learning Data Scientist
6	Advanced Data Scientist
7	Data Scientist
8	Data Scientist GenAI Agentic AI LLM RAG
9	Data Scientist AIML
10	Data Scientist/Python/MLOps- 4th Sep(Fri)-6+ y...
11	Data scientist III
12	Data Scientist
13	Data Scientist/ Data Analyst
14	Walk-in Data Scientist

Fig. 1.3: Scraped Job Records and Pandas DataFrame

The scraper collected 20 listings and extracted the required Job Title, Company Name, Location and Salary fields. The records were then converted into a structured Pandas DataFrame.

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```
7           Pune, Bengaluru Not Disclosed
8 Hybrid - Pune, Bengaluru, Delhi / NCR Not Disclosed
9           Hyderabad, Chennai, Bengaluru Not Disclosed

[18]: print("Dataset Shape:", df.shape)

      print("\nColumn Names:")
      print(df.columns.tolist())

      print("\nTotal Records:", len(df))

Dataset Shape: (20, 4)

Column Names:
['Job Title', 'Company Name', 'Location', 'Salary']

Total Records: 20

[19]: print("Salary Information:")

      print(df["Salary"].value_counts())

Salary Information:
Salary
Not Disclosed      16
20-30 Lacs PA       1
15-20 Lacs PA       1
14-20 Lacs PA       1
0-18 Lacs PA        1
Name: count, dtype: int64

[20]: df.to_csv(
      "job_market_data.csv",
```

Fig. 1.4: Dataset Structure, Salary Availability and CSV Creation

The initial dataset contains 20 records and 4 required columns. Salary information was unavailable for most listings, while a few postings contained salary ranges. The cleaned DataFrame was saved successfully as job_market_data.csv.

```
      index=False
    )
    print("CSV file saved successfully.")

CSV file saved successfully.

[21]: csv_data = pd.read_csv(
      "job_market_data.csv"
    )

    csv_data.head()

[21]:      Job Title \
0           Data Scientist
1  Data Scientist/Software Developer
2           Data Scientist III
3           Data Scientist
4 Data Scientist | Python | GenAI | LLM | RAG

      Company Name \
0 Aditya Birla Management Corporation (ABMCPL)
1           Xylem Water Solutions
2           Walmart
3           Genpact
4           Infosys

      Location      Salary
0      Bengaluru  Not Disclosed
1      Bengaluru  Not Disclosed
2      Bengaluru  Not Disclosed
3      Bengaluru  20-30 Lacs PA
```

Fig. 1.5: Verification of the Saved CSV Dataset

The generated CSV file was read back using pandas.read_csv(), and the displayed rows confirm that the job-listing data was stored correctly in

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structured form.

```
"Company Name": company,
"Location": city.title(),
"Salary": salary
})

Scraping: bangalore
Jobs found: 20

Scraping: hyderabad
Jobs found: 20

Scraping: pune
Jobs found: 20

Scraping: chennai
Jobs found: 20

Scraping: mumbai
Jobs found: 21

Scraping: delhi
Jobs found: 21

[25]: city_df = pd.DataFrame(all_jobs)

city_df = city_df.drop_duplicates().reset_index(drop=True)

print("Total jobs collected:", len(city_df))

city_df.head(10)

Total jobs collected: 60

[25]:
```

	Job Title \
0	Data Scientist
1	Data Scientist/Software Developer
2	Data Scientist III
3	Data Scientist
4	Data Scientist Python GenAI LLM RAG
5	Machine Learning Data Scientist
6	Advanced Data Scientist
7	Data Scientist

Fig. 1.6: Multi-City Job Data Collection

The same Data Scientist search was executed for Bangalore, Hyderabad, Pune, Chennai, Mumbai and Delhi. A controlled sample of 10 listings per city produced 60 records for location-wise analysis.

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```
4          Infosys Bangalore Not Disclosed
5          Hsbc Bangalore Not Disclosed
6          ExxonMobil Bangalore Not Disclosed
7          VOIS Bangalore Not Disclosed
8          Infosys Bangalore Not Disclosed
9          Tata Consultancy Services Bangalore Not Disclosed
```

```
[26]: location_analysis = (
      city_df["Location"]
      .value_counts()
      .reset_index()
      )

      location_analysis.columns = [
          "Location",
          "Number of Jobs"
      ]

      location_analysis
```

```
[26]: Location Number of Jobs
0 Bangalore 10
1 Hyderabad 10
2 Pune 10
3 Chennai 10
4 Mumbai 10
5 Delhi 10
```

```
[27]: plt.figure(figsize=(10, 5))
```

Fig. 1.7: Location-Wise Job Analysis

Pandas value_counts() produced the location-wise summary for the six-city job sample.

```
plt.tight_layout()
plt.show()
```

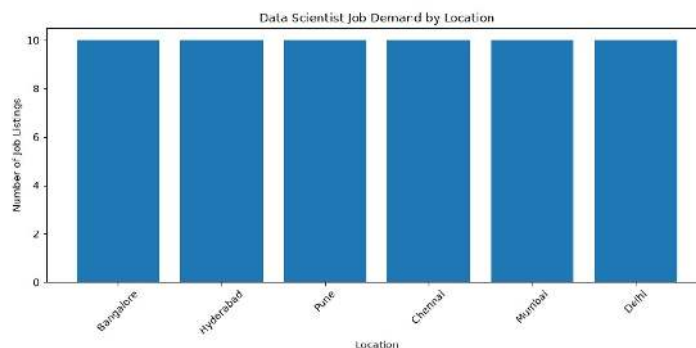


Fig. 1.8: Data Scientist Job Demand Visualization by Location

A Matplotlib bar chart visualizes the number of sampled job listings for each selected city. The graph provides a clear location-wise comparison of the collected job-market sample.

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```
[30]: print("ONLINE JOB MARKET ANALYTICS")
      print("-" * 35)

      print("Job Role:", JOB_ROLE)
      print("Total Jobs Analyzed:", len(city_df))
      print("Cities Analyzed:", len(location_analysis))

      print("\nLocation-wise Job Demand:")
      print(location_analysis.to_string(index=False))

      ONLINE JOB MARKET ANALYTICS
      -----
      Job Role: data scientist
      Total Jobs Analyzed: 60
      Cities Analyzed: 6

      Location-wise Job Demand:
      Location  Number of Jobs
      Bangalore          10

      Hyderabad          10
      Pune                10
      Chennai             10
      Mumbai              10
      Delhi               10

[31]: driver.quit()

      print("Browser closed successfully")
```

Fig. 1.9: Final Job Market Analytics Summary

The final output confirms that the multi-city dataset and graph were saved successfully. The analysis includes 60 sampled job records across 6 cities and displays the location-wise summary.

Conclusions

The exercise successfully demonstrated Selenium-based web scraping on a dynamic job portal. Job title, company name, location and available salary information were extracted from Naukri, organized in Pandas DataFrames, cleaned and stored as CSV files. Location-wise analysis was performed for six cities and visualized using a Matplotlib bar chart. Thus, the required job listings dataset, CSV output, location analysis and demand visualization were generated successfully.

References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018.
2. Python Program (GitHub link): <https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/Lab%20Codes/Week5-EP.ipynb>
3. Class lecture link: <https://drive.google.com/file/d/1HjCacfaFs49XEnuM3idduuKB0Tgjb4xI/view>